

On this page

Pulse Setup

In this page you can find a set of suggestions for Pulse monitoring using Grafana.

Suggestions



These suggestions are based on the shared resources.



There are VM resource related metrics (CPU, Memory, IO, etc) that can also be added to Grafana. These were left out on purpose Google already has out of the box monitoring support for these resources.

These suggestions can be either new dashboards, changes to existing dashboards or completely new dashboard pages. Please find the metric description in [Pulse Top Metrics](#).

Cluster page

Currently there is only one page where we can filter metrics by each of the instances. Although useful, having a cluster view monitoring page gives a view of how the overall cluster health. Given this, our suggestion is to add a new page with the following metrics:

Row Title	Title	Description	Query Details
Healthchecks - Pulse	Login	Displays the login healthcheck. This healthcheck is carried out by pulse in order to validate the login operation. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "healthcheck" = 'login' , "source" = 'pulse' , \$timeFilter Group By: time(\$__interval), "host"
Healthchecks - Pulse	Messaging	Displays the messaging healthcheck. This healthcheck is carried out by pulse in order to validate the communication with messaging server. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "healthcheck" = 'messaging' , "source" = 'pulse' , \$timeFilter Group By: time(\$__interval), "host"
Healthchecks - Pulse	Metadata DB	Displays the metadata db healthcheck. This healthcheck is carried out by pulse in order to validate the communication with metadata db. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "healthcheck" = 'metadata' , "source" = 'pulse' , \$timeFilter Group By: time(\$__interval), "host"
Healthchecks - Pulse	Zookeeper	Displays the zookeeper healthcheck. This healthcheck is carried out by pulse in order to validate the communication with zookeeper. The outcome is a bool, meaning that it	Query A Field: "result" From: "healthchecks" Where: "healthcheck" = 'zookeeper' , "source" =

Row Title	Title	Description	Query Details
		was successfully executed when 1, otherwise is 0.	'pulse' , \$timeFilter Group By: time(\$__interval), "host"
Healthchecks - Pulse	Leadership	This healthcheck indicates the leadership status of each Pulse instance, represented by 1	Query A Field: "result" From: "healthchecks" Where: "healthcheck"='leadership' , "source"='pulse' , \$timeFilter Group By: time(\$__interval), "host" Query B Field: "result" From: "healthchecks" Where: "healthcheck"='leadership' , "source"='pulse' , \$timeFilter Group By: time(\$__interval), "host"
Healthchecks - Pulse	Jobs Status	This healthcheck verifies that jobs can be submitted and ran in the Dataproc cluster. This healthcheck is based on a dummy job that is submitted every X configured time and reported in its metrics consumed by Telegraf. The outcome is a boolean, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "healthcheck" = 'job-processing-cluster' , \$timeFilter Group By: time(\$__interval), "host", "source" Query B Field: "result" From: "healthchecks" Where: "healthcheck" = 'job-processing-cluster' , \$timeFilter Group By: time(\$__interval), "host", "source"
Healthchecks - Apps	Login	Displays the appSlug login healthcheck. This healthcheck is carried out by the appSlug in order to validate the login operation. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "healthcheck"='login' , source != 'pulse' , \$timeFilter Group By: time(\$__interval), "host", "source"
Healthchecks - Apps	Leadership	This healthcheck indicates the app's leadership status, represented by 1 and 0 respectively.	Query A Field: "result" From: "healthchecks" Where: "healthcheck"='leadership' , "source"!='pulse' , \$timeFilter Group By: time(\$__interval), "source", "host" Query B Field: "result" From: "healthchecks" Where: "healthcheck"='leadership' , "source"!='pulse' , \$timeFilter Group By: time(\$__interval), "host", "source"
States	Pulse Instances State	Displays the pulse instances states: - 1: INITIAL - 2: BOOTSTRAPPING - 3: RUNNING - 4: STOPPING	Query A Field: "value" From: "pulse.server.state" Where: \$timeFilter Group By: time(\$__interval), "host"
States	Pulse Apps State	Displays the pulse appSlug states: - 1: INITIAL - 3: BOOTSTRAPPING - 4: BOOTSTRAPPED	Query A Field: "value" From: "pulse.app.state" Where: \$timeFilter

Row Title	Title	Description	Query Details
		- 5: STARTING - 6: STARTED - 8: APPLYING_CHANGES - 10: SELF_HEALING	Group By: time(\$__interval), "host", "appSlug" Query B Field: "value" From: "pulse.app.state" Where: \$timeFilter Group By: time(\$__interval), "host", "appSlug"
Leadership Changes	Pulse Leadership Changes	Measures the number of pulse leadership changes.	Query A Field: "count" From: "pulse.server.leadership.count" Where: \$timeFilter Group By: "host"
Leadership Changes	Pulse Apps Leadership Changes	Measures the number of appSlug leadership changes.	Query A Field: "count" From: "pulse.app.leadership.count" Where: \$timeFilter Group By: "host", "appSlug"
Watchdog Activations	Watchdog Activations	Measures the number watchdog activation per appSlug. Watchdog activation is triggered when there is a problem with an appSlug and subsequently, needs to be recovered.	Query A Field: "count" From: "pulse.server.watchdog.trigger" Where: \$timeFilter Group By: "host", "appSlug", "operation"
Connection Pools	Metadata DB Connection Pools Usage	Measures the metadata db connection pool usage. High values implies waiting time to access to metadata db which may degrade the component performance.	Query A Field: "value" From: "pulse.dbPool.usage" Where: appSlug = " , \$timeFilter Group By: time(\$__interval), "host", "poolName" Query B Field: "value" From: "pulse.dbPool.usage" Where: appSlug = " , \$timeFilter Group By: time(\$__interval), "host", "poolName" Query C Field: "value" From: "pulse.dbPool.maxPoolSize" Where: appSlug = " , \$timeFilter Group By: time(\$__interval), "host", "poolName"
Connection Pools	Connection Pools per App Usage	Measures the metadata connection pools usage per appSlug.	Query A Field: "value" From: "pulse.dbPool.usage" Where: appSlug != " , \$timeFilter Group By: time(\$__interval), "host", "appSlug", "poolName" Query B Field: "value" From: "pulse.dbPool.maxPoolSize" Where: appSlug != " , \$timeFilter Group By: time(\$__interval), "host", "appSlug", "poolName"
Connection Pools	Metadata DB Connection Errors and Timeouts	Measures the number of errors and timeouts occurred when trying to get a metadata db connection.	Query A Field: "count" From: "pulse.dbPool.connectionErrors" Where: appSlug = " , \$timeFilter Group By: time(\$__interval), "host", "poolName"

Row Title	Title	Description	Query Details
			Query B Field: "count" From: "pulse.dbPool.connectionObtainTimeout" Where: appSlug = " , \$timeFilter Group By: time(\$__interval), "host", "poolName" Query C Field: "count" From: "pulse.dbPool.connectionErrors" Where: appSlug = " , \$timeFilter Group By: time(\$__interval), "host", "poolName" Query D Field: "count" From: "pulse.dbPool.connectionObtainTimeout" Where: appSlug = " , \$timeFilter Group By: time(\$__interval), "host", "poolName"
Connection Pools	DB Connection Errors and Timeouts per App	Measures the number of errors and timeouts occurred when the appSlug tries to get a connection .	Query A Field: "count" From: "pulse.dbPool.connectionErrors" Where: appSlug != " , \$timeFilter Group By: time(\$__interval), "host", "poolName", "appSlug" Query B Field: "count" From: "pulse.dbPool.connectionObtainTimeout" Where: appSlug != " , \$timeFilter Group By: time(\$__interval), "host", "poolName", "appSlug" Query C Field: "count" From: "pulse.dbPool.connectionErrors" Where: appSlug != " , \$timeFilter Group By: time(\$__interval), "host", "poolName" Query D Field: "count" From: "pulse.dbPool.connectionObtainTimeout" Where: appSlug != " , \$timeFilter Group By: time(\$__interval), "host", "poolName"
Jobs	Jobs Scheduled	Measures the number of jobs scheduled for execution broken-down by job type; i.e., with state SCHEDULED (1).	Query A Field: "count" From: "pulse.job.executions.scheduled" Where: jobType = 'Workflow' , \$timeFilter Group By: time(\$__interval), *
Jobs	Jobs Waiting	Measures the number of jobs that are waiting for an agent to be free broken-down by job type; i.e., with state WAITING_AGENT (8).	Query A Field: "count" From: "pulse.job.executions.waiting" Where: \$timeFilter Group By: time(\$__interval), *
Jobs	Jobs Executing	Measures the number of jobs that are currently in state EXECUTING (2) broken-down by job type.	Query A Field: "count" From: "pulse.job.executions.running" Where: jobType = 'Workflow' , \$timeFilter Group By: time(\$__interval), *
Jobs	Jobs Failed	Measures the number of jobs that failed broken-down by both job type and	Query A Field: "count" From: "pulse.job.executions.failed"

Row Title	Title	Description	Query Details
		state: ERROR (5), CRASHED (6), ABORTED (7).	Where: jobType = 'Workflow' , \$timeFilter Group By: time(\$__interval), Query B Field: "count" From: "pulse.job.executions.failed" Where: jobType = 'Workflow' , \$timeFilter Group By: time(\$__interval),
Jobs	Jobs Finished	Measures the number of jobs that completed successfully broken-down by job type; i.e., with state FINISHED (3).	Query A Field: "count" From: "pulse.job.executions.success" Where: jobType = 'Workflow' , \$timeFilter Group By: time(\$__interval), * Query B Field: "count" From: "pulse.job.executions.success" Where: jobType = 'Workflow' , \$timeFilter Group By: time(\$__interval), "host", "appSlug"
Workflow	Global Workflow Throughput per Message Type	Measures the overall appSlug workflow throughput per message type(normal, recovery, replica).	Query A Field: "count" From: "pulse.workflow.global.event.stats" Where: \$timeFilter Group By: time(\$__interval), "host", "appSlug", "workflowDesc", "messageType"
Workflow	Global Workflow Errors	Measures the overall appSlug workflow errors, considering all message types(normal, recovery and replica).	Query A Field: "count" From: "pulse.workflow.eventErrors.count" Where: \$timeFilter Group By: time(\$__interval), "host", "appSlug", "workflowDesc"
Workflow	Global Workflow latency per Message Type at Percentile 99	Measures the overall workflow latency at percentile 99 per pulse instance, appSlug, workflow and message type.	Query A Field: "p99" From: "pulse.workflow.global.event.stats" Where: "messageType" != 'RECOVERY' , \$timeFilter Group By: time(\$__interval), "host", "appSlug", "workflowDesc", "messageType" Query B Field: "p99" From: "pulse.workflow.global.event.stats" Where: "messageType" = 'RECOVERY' , \$timeFilter Group By: time(\$__interval), "host", "appSlug", "workflowDesc" Query C Field: "p99" From: "pulse.workflow.global.event.stats" Where: "messageType" = 'NORMAL' , \$timeFilter Group By: time(\$__interval), "host", "appSlug", "workflowDesc"
Workflow	Events Spilled to Disk	Measures the overall appSlug events spilled to disk. An event is spilled to disk when pulse cannot write it on the event storage. This scenario must be avoided, otherwise require human action to move the events from the disk to the event storage.	Query A Field: "count" From: "pulse.eventStorage.spillToDisk.count" Where: \$timeFilter Group By: time(\$__interval), "host", "appSlug", "workflowDesc"

Row Title	Title	Description	Query Details
Workflow	Global Workflow Throughput by API endpoint	Measures the overall appSlug workflow throughput of normal messages by Endpoint (initiation, info, etc).	Query B Field: "count" From: "workflow.events" Where: \$timeFilter Group By: time(\$__interval), "workflowDesc", "elementDesc", "host", "appSlug", "event_type"
Event Storage Dumps	Events Storage Dump Failed and Retry Events	Measures the overall number of events that were dumped to disk and are either pending to be retried or reached to the max amount of retries and need to be re-injected manually.	Query A Field: "count" From: "pulse.dump.storage.dump.entries.retry.counter" Where: \$timeFilter Group By: time(\$__interval), "host", "appSlug", "workflowDesc" Query B Field: "count" From: "pulse.dump.storage.dump.entries.failed.counter" Where: \$timeFilter Group By: time(\$__interval), "host", "appSlug", "workflowDesc"
Event Storage Dumps	Events Storage Dump Failed and Retry Queue Size	Measures the overall number of events that are present in the in-memory queue to be written to the dump file in the failed folder, appSlug and workflow.	Query A Field: "value" From: "pulse.dump.storage.failed.queue.size" Where: \$timeFilter Group By: time(\$__interval), "host", "appSlug", "workflowDesc" Query B Field: "value" From: "pulse.dump.storage.retry.queue.size" Where: \$timeFilter Group By: time(\$__interval), "host", "appSlug", "workflowDesc"
Event Pending Dumps	Event Pending Persistence to Event Storage	Measures the overall number of events that are pending to be persisted to the databased.	Query A Field: "count" From: "pulse.workflow.eventstorage.eventsQueuedSize" Where: \$timeFilter Group By: time(\$__interval), "host", "appSlug", "workflowDesc"
Profile Fetches	Profile Fetches Latency at Percentile 99 (Excluding Timeouts)	Measures the overall max latency to fetch the profiles at percentile 99, per appSlug, workflow and plan.	Query A Field: "p99" From: "pulse.workflow.profileFetches.statistics" Where: \$timeFilter Group By: time(\$__interval), "host", "appSlug", "workflowDesc", "elementDesc"
Profile Fetches	Profile Fetches Timeouts	Measures the number of profile fetches that were timed out, per appSlug, workflow and plan.	Query A Field: "count" From: "pulse.workflow.profileFetches.timeouts" Where: \$timeFilter Group By: time(\$__interval), "host", "appSlug", "workflowDesc", "elementDesc"
JVM	JVM Heap Usage (Pulse Server)	Measures the jvm heap usage per pulse instance.	Query A Field: "value" From: "letmeknow.jvm.memory.heap.usage" Where: appSlug = " , \$timeFilter Group By: time(\$__interval), "host"

Row Title	Title	Description	Query Details
			Query B Field: "value" From: "letmeknow.jvm.memory.heap.usage" Where: appSlug = " , \$timeFilter Group By: time(\$__interval), "host"
JVM	JVM Heap Usage (per App)	Measures the jvm heap usage per application and host.	Query A Field: "value" From: "letmeknow.jvm.memory.heap.usage" Where: appSlug != " , \$timeFilter Group By: time(\$__interval), "host", "appSlug" Query B Field: "value" From: "letmeknow.jvm.memory.heap.usage" Where: appSlug != " , \$timeFilter Group By: time(\$__interval), "host", "appSlug" Query C Field: "value" From: "letmeknow.jvm.memory.heap.usage" Where: appSlug != " , \$timeFilter Group By: time(\$__interval), "host", "appSlug"
JVM	JVM Heap Usage After GC (Pulse Server)	Measures the jvm heap usage after gc per pulse instance.	Query A Field: "value" From: "letmeknow.jvm.memory.heap.after-gc.usage" Where: appSlug = " , \$timeFilter Group By: time(\$__interval), "host"
JVM	JVM Heap Usage After GC (per App)	Measures the jvm heap usage after gc per application and host.	Query A Field: "value" From: "letmeknow.jvm.memory.heap.after-gc.usage" Where: appSlug != " , \$timeFilter Group By: time(\$__interval), "host", "appSlug"
JVM	JVM Code Cache Usage (Pulse Server)	Measures the jvm code cache usage per pulse instance.	Query A Field: "value" From: "letmeknow.jvm.memory.pools.CodeCache.usage" Where: appSlug = " , \$timeFilter Group By: time(\$__interval), "host"
JVM	JVM Code Cache Usage (per App)	Measures the jvm code cache usage per application and host.	Query A Field: "value" From: "letmeknow.jvm.memory.pools.CodeCache.usage" Where: appSlug != " , \$timeFilter Group By: time(\$__interval), "host", "appSlug"
JVM	JVM Garbage Collector Time (Pulse Server)	Measures maximum JVM GC duration. Long garbage collection pause durations indicate that the heap size is not adjusted for the actual usage.	Query A Field: "value" From: "letmeknow.jvm.gc.G1-Old-Generation.time" Where: appSlug = " , \$timeFilter Group By: "host" Query B Field: "value" From: "letmeknow.jvm.gc.G1-Young-Generation.time" Where: appSlug = " , \$timeFilter Group By: "host"
JVM	JVM Garbage Collector Time (per App)	Measures maximum JVM GC duration for young and old gen, per application and host. Long garbage collection	Query A Field: "value" From: "letmeknow.jvm.gc.G1-Old-Generation.time"

Row Title	Title	Description	Query Details
		pause durations indicate that the heap size is not adjusted for the actual usage.	Where: appSlug != " , \$timeFilter Group By: "host", "appSlug" Query B Field: "value" From: "letmeknow.jvm.gc.G1-Young-Generation.time" Where: appSlug != " , \$timeFilter Group By: "host", "appSlug"
JVM	JVM Active Threads Count (Pulse Server)	Measures the current number of active threads in the JVM per pulse instance.	Query A Field: "value" From: "letmeknow.jvm.threads.count" Where: appSlug = " , \$timeFilter Group By: time(\$__interval), "host"
JVM	JVM Active Threads Count (per App)	Measures the current number of active threads in the JVM per application and host.	Query A Field: "value" From: "letmeknow.jvm.threads.count" Where: appSlug != " , \$timeFilter Group By: time(\$__interval), "host", "appSlug"
JVM	JVM File Descriptors Ratio (Pulse Server)	Measures the jvm file descriptors usage ratio per instance.	Query A Field: "value" From: "letmeknow.jvm.filedescriptor" Where: appSlug = " , \$timeFilter Group By: time(\$__interval), "host"
JVM	JVM File Descriptors Ratio (per App)	Measures the jvm file descriptors usage ratio per application and host.	Query A Field: "value" From: "letmeknow.jvm.filedescriptor" Where: appSlug != " , \$timeFilter Group By: time(\$__interval), "host", "appSlug"

Note that in the cluster page some dashboards are separated. For instance, instead of having the **Jobs** dashboards separated by job state, we can have a dashboard that gathers all states. Given that in cluster view we are not filtering by any node, it is recommend to have disassociated dashboards so they do not get cluttered with information.

Also note the **Row Title** column which refers to a Grafana feature to group dashboards by type. This is only a suggestion and thus not mandatory.

Node page

These suggestions relate to the existing Pulse dashboard page. You may find some new dashboards or changes to existing ones.

Our suggestions are to add the dashboards in the following table with a filter by each of the available nodes and applications:

Row Title	Title	Description	Query Details
Healthchecks - Pulse	Login	Displays the login healthcheck on the selected instance. This healthcheck is carried out by pulse in order to validate the login operation. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "source" = 'pulse' , "healthcheck" = 'login' , "host" = '\$host' , \$timeFilter Group By: time(\$__interval)

Row Title	Title	Description	Query Details
Healthchecks - Pulse	Messaging	Displays the messaging healthcheck on the selected instance. This healthcheck is carried out by pulse in order to validate the communication with rabbitmq. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "source" = 'pulse' , "healthcheck" = 'messaging' , "host" = '\$host' , \$timeFilter Group By: time(\$__interval)
Healthchecks - Pulse	Metadata DB	Displays the metadata db healthcheck on the selected instance. This healthcheck is carried out by pulse in order to validate the communication with metadata db. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "source" = 'pulse' , "healthcheck" = 'metadata' , "host" = '\$host' , \$timeFilter Group By: time(\$__interval)
Healthchecks - Pulse	Zookeeper	Displays the zookeeper healthcheck on the selected instance. This healthcheck is carried out by pulse in order to validate the communication with zookeeper. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "source" = 'pulse' , "healthcheck" = 'zookeeper' , "host" = '\$host' , \$timeFilter Group By: time(\$__interval)
Pulse State and Leadership Changes	Pulse Leadership Changes	Displays pulse leadership changes, for the selected pulse instance.	Query A Field: "count" From: "pulse.server.leadership.count" Where: "host" = '\$host' , \$timeFilter Group By: N/A
Healthchecks - Apps	Login	Displays the login healthcheck for the specified app on the selected instance. This healthcheck is carried out by the appSlug in order to validate the login operation. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "healthcheck"='login' , source =~ /^app.\$appSlug.* / , "host" = '\$host' , \$timeFilter Group By: time(\$__interval)
Healthchecks - Apps	Leadership	Displays the leadership healthcheck for the specified app on the selected instance. This healthcheck is carried out by the appSlug in order to validate the login operation. The outcome is a bool, meaning that it was successfully executed when 1, otherwise is 0.	Query A Field: "result" From: "healthchecks" Where: "healthcheck"='leadership' , source =~ /^app.\$appSlug.* / , "host" = '\$host' , \$timeFilter Group By: time(\$__interval)
Watchdog Activations	Watchdog Activations	Measures the number of watchdog activations triggered for the selected appSlug and the specified pulse instance.	Query A Field: "count" From: "pulse.server.watchdog.trigger" Where: "host" = '\$host' , "appSlug" = '\$appSlug' , \$timeFilter Group By: "operation"
Jobs	Jobs Scheduled	Measures the number of jobs scheduled for execution broken-down by job type; i.e., with state SCHEDULED (1).	Query A Field: "count" From: "pulse.job.executions.scheduled" Where: jobType = 'Workflow' , "host" = '\$host' , "appSlug" = '\$appSlug' , \$timeFilter Group By: time(\$__interval), "host", "appSlug", "jobType"
Jobs	Jobs Waiting	Measures the number of jobs that are waiting for an agent to be free broken-down by job type; i.e., with state WAITING_AGENT (8).	Query A Field: "count" From: "pulse.job.executions.waiting" Where: jobType = 'Workflow' , "host" = '\$host' ,

Row Title	Title	Description	Query Details
			"appSlug" = '\$appSlug' , \$timeFilter Group By: time(\$__interval), "host", "appSlug", "jobType"
Jobs	Jobs Executing	Measures the number of jobs that are currently in state EXECUTING (2) broken-down by job type.	Query A Field: "count" From: "pulse.job.executions.running" Where: jobType = 'Workflow' , "host" = '\$host' , "appSlug" = '\$appSlug' , \$timeFilter Group By: time(\$__interval), "host", "appSlug", "jobType"
Jobs	Jobs Failed	Measures the number of jobs that failed broken-down by both job type and state: ERROR (5), CRASHED (6), ABORTED (7).	Query A Field: "count" From: "pulse.job.executions.failed" Where: jobType = 'Workflow' , "host" = '\$host' , "appSlug" = '\$appSlug' , \$timeFilter Group By: time(\$__interval), "host", "appSlug", "jobType"
Jobs	Jobs Finished	Measures the number of jobs that completed successfully broken-down by job type; i.e., with state FINIDHED (3).	Query A Field: "count" From: "pulse.job.executions.success" Where: jobType = 'Workflow' , "host" = '\$host' , "appSlug" = '\$appSlug' , \$timeFilter Group By: time(\$__interval), "host", "appSlug", "jobType"